

Week 7 Cost-Benefit Memo

Verdict

Although the Random Forest model achieved better overall predictive performance than the Logistic Regression baseline, I recommend retaining the Logistic Regression model for Phase 3 until further optimisation demonstrates that a more complex model can improve overall performance without reducing its ability to identify the most critically ill patients.

Dataset and Methods

This analysis used the cleaned Yale Emergency Medicine Machine Learning Challenge (Yale EMMLC) emergency department triage dataset previously prepared during Weeks 5 and 6. The same preprocessing pipeline and 80/20 train-test split from Week 6 were reused to ensure a fair comparison between models.

The Week 6 Logistic Regression model served as the baseline. A Random Forest classifier was implemented as the more complex model and evaluated using the same test dataset. Hyperparameter optimisation was performed using Randomized SearchCV before evaluating the final model.

Performance was assessed using seven evaluation criteria:

1. Accuracy
2. Macro Precision
3. Macro Recall
4. Macro F1-score
5. Training time
6. Inference time
7. Interpretability

Because the application is emergency department triage, the models were also compared on their ability to identify the highest-acuity patients (ESI Level 1).

Benchmark Results

Evaluation Criterion	Logistic Regression (Week 6 Baseline)	Tuned Random Forest (Week 7)	Better Model
Accuracy	0.5629	0.6042	Random Forest
Macro Precision	0.4131	0.4773	Random Forest
Macro Recall	0.6166	0.5438	Logistic Regression
Macro F1-score	0.4119	0.4918	Random Forest
Training Time (s)	15.62	22.18	Logistic Regression
Inference Time (ms/prediction)	0.001426	0.074344	Logistic Regression
Interpretability	High	Moderate	Logistic Regression

Benefits of the Recommended Model

1. Better clinical transparency

Logistic Regression remains straightforward to interpret. Individual model coefficients can be explained to clinicians, allowing emergency physicians to understand why particular predictions are made. This transparency encourages clinician trust and supports safe clinical decision-making.

2. Lower computational cost

The baseline model trained faster than the Random Forest and required substantially less computation during prediction. Although both inference times are fast, the simpler model places less demand on hospital computing resources and is easier to maintain.

3. Better identification of the sickest patients

While Random Forest achieved higher overall accuracy, Logistic Regression achieved substantially higher recall for ESI Level 1 patients (0.625 versus 0.312). Missing critically ill patients carries greater clinical risk than small improvements in overall classification performance, making this an important consideration for emergency department deployment.

Arguments Against the Recommendation

1. Lower overall predictive performance

Random Forest produced higher overall accuracy, macro precision and macro F1-score, indicating stronger classification performance across the dataset.

2. Simpler decision boundaries

Logistic Regression assumes largely linear relationships between variables. Random Forest can model more complex interactions between physiological measurements that Logistic Regression may not capture.

3. Potential future improvements

Further optimisation of Random Forest or exploration of other ensemble methods such as Gradient Boosting or XGBoost may improve critical patient detection while maintaining superior overall accuracy.

Risks and Unknowns

Several uncertainties remain before deployment.

The dataset represents a single source and may not fully reflect Caribbean emergency department populations. External validation is therefore required before any production implementation.

Although Random Forest improved several evaluation metrics, its poorer performance on ESI Level 1 patients requires further investigation. Additional feature engineering, hyperparameter optimisation, or class balancing techniques may improve this weakness.

Finally, this evaluation focused on offline model performance. Workflow integration, clinician acceptance, maintenance requirements and ongoing model monitoring have not yet been assessed.

Recommendation

At this stage, Logistic Regression should remain the preferred model for Phase 3 because it provides strong interpretability, lower computational cost and better identification of critically ill patients. The Random Forest model should continue to be investigated as a promising alternative due to its superior overall predictive performance, but additional optimisation is required before it can be recommended for deployment in a clinical environment.